

ALFIDO TECH INTERNSHIP PROGRAM

COHORT: MAY–JUNE 2026

- **Candidate Name:** Manish Kafle
- **Candidate ID:** BS/REG/120773
- **Domain:** Python Developer Intern
- **Project Task:** Task 3 – Data Analysis with Pandas

1. Project Objective

The objective of this project is to implement an automated data analysis pipeline using the Python pandas library to audit, clean, and derive actionable academic insights from raw student datasets. The application structurally imports an external CSV database registry, applies computational imputation to handle data gaps safely, filters high-performance student tiers via conditional indexing, and generates aggregate department performance metrics.

2. Technical Stack Used

- **Core Language Client:** Python 3.x
- **Core Analysis Engine:** Pandas
- **Execution Environment:** Jupyter Notebook
- **Data Format Structure:** Comma-Separated Values

3. Repository Links

- **Task Folder Repository Link:** <https://github.com/CodedByManish/Alfido-python-internship/tree/main/task-03-student-analysis>

4. Notebook Execution Cells & Verification Outputs

Documentation Note: Since this analysis was performed inside an interactive Jupyter Notebook (.ipynb) environment, the core source code cells and their respective terminal execution outputs are captured together in the single verification screen states below.

Phase 1: Dataset Ingestion & Structural Inspection (Cells 1 & 2)

- **Applied Logic:** The program initializes the storage array from `students.csv`. Running `df.info()` and `df.isnull().sum()` outputs the data schema shape, isolating missing data matrices within the column index architecture before starting structural mutations.

```
import pandas as pd

# Load dataset
df = pd.read_csv("students.csv")
print(f"Dataset loaded.")
```

✓ 0.0s

Dataset loaded.

```
# Check structure and missing values
df.info()

print("\nMissing values:")
print(df.isnull().sum())
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Name                   20 non-null    str
1   Department             20 non-null    str
2   Age                    20 non-null    int64
3   Gender                 20 non-null    str
4   Marks                  19 non-null    float64
5   Attendance(%)          20 non-null    int64
dtypes: float64(1), int64(2), str(3)
memory usage: 1.1 KB
```

Missing values:

```
Name           0
Department      0
Age             0
Gender          0
Marks           1
Attendance(%)   0
dtype: int64
```

Phase 2: Data Cleansing & Statistical Imputation (Cell 3)

- **Applied Logic:** Missing entries found inside the Marks vector are dynamically replaced with the calculated dataset mean using pandas `.fillna()`. This preserves rows with valid neighboring parameters and ensures final calculations aren't skewed.

```
# Fill missing marks with mean value
mean_marks = df["Marks"].mean()
df["Marks"] = df["Marks"].fillna(mean_marks)

print(f"Missing values filled with mean: {round(mean_marks, 1)}")

df
```

✓ 0.0s

Missing values filled with mean: 84.0

	Name	Department	Age	Gender	Marks	Attendance(%)
0	Manish	IT	21	M	85.0	92
1	Sita	IT	20	F	90.0	95
2	Ram	BCA	22	M	78.0	88
3	Hari	BCA	21	M	84.0	80
4	Gita	CSIT	20	F	88.0	94
5	Nabin	IT	22	M	95.0	97
6	Priya	CSIT	21	F	82.0	89
7	Kiran	BCA	20	M	76.0	85
8	Anita	IT	21	F	91.0	96
9	Rohit	CSIT	22	M	89.0	90
10	Aarav	IT	20	M	73.0	75
11	Asha	BCA	21	F	84.0	87
12	Binod	CSIT	22	M	79.0	83
13	Sunita	IT	20	F	92.0	98
14	Sagar	BCA	23	M	81.0	86
15	Maya	CSIT	21	F	87.0	91
16	Dipak	IT	22	M	69.0	72
17	Laxmi	BCA	20	F	94.0	99
18	Anil	CSIT	21	M	77.0	80
19	Pooja	IT	20	F	86.0	93

Phase 3: Performance Segmentation & Group Analytics

Applied Logic: This final phase uses conditional boolean masking to filter out top-performing students (Marks > 90) and groups categorical department profiles using `.groupby()`. Running `.describe()` summarizes the clean collection's distribution data by printing percentiles, standard deviations, and core mean tracking figures.

```
import pandas as pd

# Load dataset
df = pd.read_csv("students.csv")
print(f"Dataset loaded.")
```

✓ 0.0s

Dataset loaded.

```
# Check structure and missing values
df.info()

print("\nMissing values:")
print(df.isnull().sum())
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
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0   Name                   20 non-null    str
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3   Gender                  20 non-null    str
4   Marks                   19 non-null    float64
5   Attendance(%)          20 non-null    int64
dtypes: float64(1), int64(2), str(3)
memory usage: 1.1 KB
```

```
Missing values:
Name          0
Department    0
Age           0
Gender        0
Marks         1
Attendance(%) 0
dtype: int64
```